Constant Volume Heating with Nitrogen blanketing

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1 Introduction

During one of projects I and other engineers were tasked with sizing pressure relief deice (PRD). For two phase relief of foamy fluids we were told to use ChemCad package FERST Fauske. We had internal procedures to follow to calculate "conservaitve" results (I won't expand on why it's in quotes), but there were some problems with finding the right liquid swell. The software sometimes gave the error that there is more liquid in the tank than the tank's volume itself. This send me on a hunt for solution as no one could answer my simple question: why?

There is a great deal of articles from people associated with DIERS and Fauske Association that were mentioning one thing over an over again, there is a lower bound on the maximum liquid level in the tank above which no matter if fluid is foamy or not, we should size the PRD for two phase flow. This means that liquid swell is a very important parameter to get right in order to size any PRD correctly.

2 Assumptions

We start with a tank of volume V that is filled with liquid to some level λ with the rest of the tanks volume being filled by nitrogen blanketing. If all of inlet/outlet valves are closed and we now provide external heating (fire for example) then we are dealing with isochoric heat addition process.

Let us also make few assumptions needed:

1. Tank volume is constant and material its made of is non elastic.
2. Nitrogen and liquid vapors can be treated as ideal gas.
3. Amagat's and Rault's laws are applicable.

4. Liquid in tank is incompressible.
5. Nitrogen is insoluble in liquid.
6. Liquid, vapor and nitrogen are in mechanical and thermal equillibrium.
7. Liquid and vapor are in phase equillibrium at each iteration step.
8. Mixture in the tank can be treated as ideal solution.

3 Derivation

3.1 Basics

We have already stated, that the process is isochoric in nature. Mathematically:

$$dV = 0$$

Because of our assumptions we can write that the total volume is a sum of all the volumes of the system, thus:

$$dV = dV_{liquid} + dV_{gases}$$

where we will write V_l and V_g for short. By Amagat's law we have that:

$$dV_g = dV_{N_2} + dV_v$$

and so, to sum it up, our system must follow:

$$dV = dV_l + dV_v + dV_{N_2} = 0$$

Now, we can find each component differential volume change in (T, P) plug it in and solve. For each component volume we have that $V_i = m_i v_i$, and so we can analyze it further:

$$dV_i = m_i dv_i + v_i dm_i$$

This immediatelly tells us that the change in volume of components is due to change in specific volume but also in mass change, which will be true only for liquid and it's vapor, but nitrogen mass is constant throughout the process.

We now write that the total differential for any component dv_i is:

$$dv_i = \left(\frac{\partial v_i}{\partial T}\right)_{P,mf_j} dT + \left(\frac{\partial v_i}{\partial P}\right)_{T,mf_j} dP + \left(\frac{\partial v_i}{\partial mf_i}\right)_{T,P,mf_j} dmf_i$$

Because of assumption (8), we can drop partial specific volume and have this headache dealt with right out of the gate. For the mass total differential we can use the property that sum of liquid and vapor must be equal to total fluid mass that we filled the tank with. In other words, because $m = m_l + m_v$, we can write,

$$dm_l = -dm_v$$

and

$$dm_v = \left(\frac{\partial m_v}{\partial T}\right)_P dT + \left(\frac{\partial m_v}{\partial P}\right)_T dP$$

We will start from the easiest of differentials, a nitrogen one.

3.2 Nitrogen volume differential

For the nitrogen, its mass is held constant during the process and so the volume change is related only to specific volume change. Because of assumption (2) we can immediatelly write that:

$$dV_{N_2} = m_{N_2} dv_{N_2} = m_{N_2} \left(\frac{R_{N_2}}{P} dT - \frac{R_{N_2} T}{P^2} dP \right)$$

and that's it.

3.3 Vapor volume differential

3.3.1 Vapor mass differential dm_v

Because of assumptions (3), (7) and (8) we can write that the molar fraction of vapor in nitrogen blanketing is:

$$y_v = \frac{P_{sat}(T)}{P}$$

and because we are dealing with only nitrogen and vapor $y_{N_2} = 1 - y_v$ which gives us that the mass fraction is:

$$mf_v = \frac{y_v M_v}{M_{N_2} + (M_v - M_{N_2}) y_v}$$

Because nitrogen mass is related to total gas mixture by $m_{N_2} = mf_{N_2} \cdot (m_v + m_{N_2})$ we can finally write that vapor mass is:

$$m_v = \frac{mf_v}{1 - mf_v} m_{N_2}$$

We will be, in the end, solving it all iteratively and so there is no reason to find exact partial derivatives.

3.3.2 Vapos spec. volume differential

We haveve assumed that we can treat vapor as ideal gas by (2) which gives us similar solution like in nitrogen case, and so we have:

$$dv_v = \frac{R_v}{P} dT - \frac{R_v T}{P^2} dP$$

3.4 Liquid volume differential

We write that total differential of specific volume of liquid is:

$$dv_l = v_l \beta dT - v_l \alpha dP$$

and because of (4) we can drop pressure dependency here. We have already derived its mass differential in (3.3.1). Basically, we can start to construct parametric curve for the process.

4 Process path

We will now list all volume differentials one by one.

$$dV_l = \left[(m - m_v)v_l\beta - v_l\left(\frac{\partial m_v}{\partial T}\right)_P\right]dT - v_l\left(\frac{\partial m_v}{\partial P}\right)_T dP$$

$$dV_v = \frac{R_v}{P}\left[m_v + T\left(\frac{\partial m_v}{\partial T}\right)_P\right]dT + \frac{R_v T}{P}\left[\left(\frac{\partial m_v}{\partial P}\right)_T - \frac{m_v}{P}\right]dP$$

$$dV_{N_2} = m_{N_2}\left(\frac{R_{N_2}}{P}dT - \frac{R_{N_2}T}{P^2}dP\right)$$

5 References